

Entity, Attribute, Key and Relationship Identification

E-Commerce Order Management Database System

Field	Details
Course	Database Management Systems (Skill Development Activity)
Task	Task 2: Identification of Entities, Attributes, Primary Keys and Relationships
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1. Introduction

This document is the outcome of the data requirement study of the E-Commerce Order Management System. Based on the requirement specification (Task 1), it identifies the entities that must be stored, their attributes, the primary key of each entity, the foreign keys that connect them, and the relationships between entities with their cardinality. It is the input for drawing the Entity Relationship (ER) diagram and the relational schema.

1.1 Approach Followed

1. Read the business scenario and underline the nouns that represent things the business needs to store (candidate entities).
2. List the facts that describe each entity (attributes).
3. Choose an attribute that uniquely identifies each record (primary key).
4. Study the verbs that connect the entities (relationships) and decide their cardinality.
5. Resolve many-to-many relationships using associative entities.

1.2 Concepts Used

Term	Meaning
Entity	A real-world object about which data is stored (for example, Customer)
Attribute	A property that describes an entity (for example, Email)
Primary Key (PK)	An attribute that uniquely identifies each record and can never be NULL
Candidate / Alternate Key	An attribute that could also uniquely identify records but was not chosen as the PK
Foreign Key (FK)	An attribute that refers to the primary key of another entity to link them
Relationship	An association between two or more entities
Cardinality	The number of records of one entity that can relate to records of another (1:1, 1:N, M:N)
Participation	Whether every record must take part in the relationship (total) or not (partial)

2. Identification of Entities

From the business scenario, nine entities are identified.

No.	Entity	Type	Description
1	Customer	Strong	A person registered on the platform who buys products
2	Category	Strong	A group under which products are classified (for example, Electronics)
3	Supplier	Strong	A vendor or company that supplies products
4	Product	Strong	An item available for sale, with its price and stock
5	Order	Strong	A purchase request placed by a customer
6	Order Details	Associative (weak)	The individual products and quantities that belong to an order
7	Payment	Dependent on Order	The payment made for an order
8	Shipment	Dependent on Order	The delivery of an order to the customer
9	Review	Associative	The rating and comments given by a customer for a product

Note on naming: ORDER is a reserved keyword in SQL (used in ORDER BY). In the implementation, the Order entity will therefore be created as a table named ORDERS.

3. Attributes and Primary Keys of Each Entity

In the tables below, PK means primary key, FK means foreign key, UK means unique key, and NN means NOT NULL. Suggested data types are given to help in the later SQL implementation.

3.1 Customer

Attribute	Data Type	Key / Constraint	Description
Customer ID	NUMBER	PK	Unique identifier of the customer
Name	VARCHAR(100)	NN	Full name
Email	VARCHAR(100)	UK, NN	Email address (candidate key)
Mobile Number	VARCHAR(15)	UK, NN	Contact number (candidate key)
Address	VARCHAR(255)	NN	Residential address
Registration Date	DATE	Default: current date	Date on which the customer registered

Primary key: Customer ID. **Candidate keys:** Email, Mobile Number. Customer ID is chosen because it is short, never changes, and is system generated.

3.2 Category

Attribute	Data Type	Key / Constraint	Description
Category ID	NUMBER	PK	Unique identifier of the category
Category Name	VARCHAR(100)	UK, NN	Name of the category
Description	VARCHAR(255)	Optional	Short description of the category

Primary key: Category ID. **Candidate key:** Category Name.

3.3 Supplier

Attribute	Data Type	Key / Constraint	Description
Supplier ID	NUMBER	PK	Unique identifier of the supplier
Supplier Name	VARCHAR(100)	NN	Name of the supplier
Contact Information	VARCHAR(150)	NN	Phone number, email or address of the supplier

Primary key: Supplier ID.

3.4 Product

Attribute	Data Type	Key / Constraint	Description
Product ID	NUMBER	PK	Unique identifier of the product
Product Name	VARCHAR(150)	NN	Name of the product
Price	NUMBER(10,2)	NN, CHECK > 0	Current selling price
Stock Quantity	NUMBER	NN, CHECK >= 0, Default 0	Units currently available
Category ID	NUMBER	FK to Category	Category to which the product belongs
Supplier ID	NUMBER	FK to Supplier	Supplier who provides the product

Primary key: Product ID. **Foreign keys:** Category ID, Supplier ID. **Design note:** The given entity list does not show how Supplier is linked to Product. Supplier ID has been added to Product as a foreign key so that the Supplier entity is connected to the rest of the model.

3.5 Order

Attribute	Data Type	Key / Constraint	Description
Order ID	NUMBER	PK	Unique identifier of the order
Customer ID	NUMBER	FK to Customer, NN	Customer who placed the order
Order Date	DATE	Default: current date	Date on which the order was placed
Total Amount	NUMBER(12,2)	CHECK >= 0	Total value of the order
Order Status	VARCHAR(20)	CHECK in (Placed, Confirmed, Shipped, Delivered, Cancelled)	Current state of the order

Primary key: Order ID. **Foreign key:** Customer ID.

3.6 Order Details

Attribute	Data Type	Key / Constraint	Description
Order Detail ID	NUMBER	PK	Unique identifier of the line item
Order ID	NUMBER	FK to Order, NN	Order to which the item belongs
Product ID	NUMBER	FK to Product, NN	Product that was ordered
Quantity	NUMBER	NN, CHECK > 0	Number of units ordered
Unit Price	NUMBER(10,2)	NN, CHECK > 0	Price of one unit at the time of purchase

Primary key: Order Detail ID. **Foreign keys:** Order ID, Product ID. The combination (Order ID, Product ID) is a candidate key because the same product should appear only once in an order.

3.7 Payment

Attribute	Data Type	Key / Constraint	Description
Payment ID	NUMBER	PK	Unique identifier of the payment
Order ID	NUMBER	FK to Order, NN	Order for which the payment is made
Payment Method	VARCHAR(30)	CHECK in (UPI, Credit Card, Debit Card, Net Banking, Cash on Delivery)	Mode of payment
Payment Date	DATE	Default: current date	Date of the payment
Payment Status	VARCHAR(20)	CHECK in (Pending, Completed, Failed, Refunded)	Result of the payment

Primary key: Payment ID. **Foreign key:** Order ID.

3.8 Shipment

Attribute	Data Type	Key / Constraint	Description
Shipment ID	NUMBER	PK	Unique identifier of the shipment
Order ID	NUMBER	FK to Order, UK, NN	Order being delivered
Delivery Address	VARCHAR(255)	NN	Address to which the order is delivered
Shipment Date	DATE	Optional until dispatched	Date of dispatch
Delivery Status	VARCHAR(20)	CHECK in (Processing, Shipped, Out for Delivery, Delivered, Returned)	Current delivery state

Primary key: Shipment ID. **Foreign key:** Order ID (unique, because each order has one shipment).

3.9 Review

Attribute	Data Type	Key / Constraint	Description
Review ID	NUMBER	PK	Unique identifier of the review
Customer ID	NUMBER	FK to Customer, NN	Customer who wrote the review
Product ID	NUMBER	FK to Product, NN	Product being reviewed
Rating	NUMBER(1)	NN, CHECK between 1 and 5	Star rating given
Comments	VARCHAR(500)	Optional	Written feedback

Primary key: Review ID. **Foreign keys:** Customer ID, Product ID. The combination (Customer ID, Product ID) is a candidate key because a customer reviews a product only once.

4. Summary of Primary Keys and Foreign Keys

Entity	Primary Key	Foreign Keys	Refers to
Customer	Customer ID	None	None
Category	Category ID	None	None
Supplier	Supplier ID	None	None
Product	Product ID	Category ID, Supplier ID	Category, Supplier
Order	Order ID	Customer ID	Customer

Entity	Primary Key	Foreign Keys	Refers to
Order Details	Order Detail ID	Order ID, Product ID	Order, Product
Payment	Payment ID	Order ID	Order
Shipment	Shipment ID	Order ID	Order
Review	Review ID	Customer ID, Product ID	Customer, Product

5. Identification of Relationships

5.1 Relationship Details

No.	Relationship	Cardinality	Participation	FK placed in
R1	Customer places Order	1:N	Customer: partial; Order: total	Order (Customer ID)
R2	Category contains Product	1:N	Category: partial; Product: total	Product (Category ID)
R3	Supplier supplies Product	1:N	Supplier: partial; Product: total	Product (Supplier ID)
R4	Order includes Product	M:N	Order: total; Product: partial	Resolved by Order Details
R5	Order has Order Details	1:N	Order: total; Order Details: total	Order Details (Order ID)
R6	Product appears in Order Details	1:N	Product: partial; Order Details: total	Order Details (Product ID)
R7	Order is paid by Payment	1:1	Order: partial; Payment: total	Payment (Order ID)
R8	Order is delivered by Shipment	1:1	Order: partial; Shipment: total	Shipment (Order ID, unique)
R9	Customer writes Review	1:N	Customer: partial; Review: total	Review (Customer ID)
R10	Product receives Review	1:N	Product: partial; Review: total	Review (Product ID)
R11	Customer reviews Product	M:N	Both partial	Resolved by Review

5.2 One-to-One Relationships

- **Order to Payment:** in the basic design each order has one payment record. If the business later allows partial payments or payment retries, this becomes one-to-many.
- **Order to Shipment:** each order is shipped as a single shipment, and each shipment belongs to one order. The unique constraint on Shipment.Order ID enforces this.

5.3 One-to-Many Relationships

- One **Customer** can place many **Orders**.
- One **Category** can contain many **Products**.
- One **Supplier** can supply many **Products**.
- One **Order** can contain many **Order Details** (line items).

5.4 Many-to-Many Relationships and How They Are Resolved

A many-to-many relationship cannot be stored directly in a relational table, so it is broken into two one-to-many relationships using an associative (junction) table.

Many-to-Many Relationship	Associative Entity	Resulting One-to-Many Relationships
Order and Product: an order has many products, and a product appears in many orders	Order Details	Order to Order Details, and Product to Order Details

Many-to-Many Relationship	Associative Entity	Resulting One-to-Many Relationships
Customer and Product (reviews): a customer reviews many products, and a product is reviewed by many customers	Review	Customer to Review, and Product to Review

Order Details also carries its own attributes (Quantity and Unit Price) and Review carries Rating and Comments, which is why these relationships are best modelled as entities.

5.5 Relationship Matrix

	Cust.	Categ.	Suppl.	Prod.	Order	Ord. Det.	Paym.	Ship.	Rev.
Customer	-	-	-	M:N	1:N	-	-	-	1:N
Category	-	-	-	1:N	-	-	-	-	-
Supplier	-	-	-	1:N	-	-	-	-	-
Product	M:N	N:1	N:1	-	M:N	1:N	-	-	1:N
Order	N:1	-	-	M:N	-	1:N	1:1	1:1	-
Order Details	-	-	-	N:1	N:1	-	-	-	-
Payment	-	-	-	-	1:1	-	-	-	-
Shipment	-	-	-	-	1:1	-	-	-	-
Review	N:1	-	-	N:1	-	-	-	-	-

6. Textual ER Overview

The following notation summarizes how the entities connect (1 = one, N = many):

- Customer (1) places (N) Order
- Order (1) contains (N) Order Details, and Product (1) is ordered in (N) Order Details
- Order (1) is paid by (1) Payment
- Order (1) is delivered by (1) Shipment
- Category (1) groups (N) Product
- Supplier (1) supplies (N) Product
- Customer (1) writes (N) Review, and Product (1) is reviewed in (N) Review

This overview is the basis for the complete ER diagram prepared in the next task.

7. Sample Records (for Validation of the Design)

Customer

Customer ID	Name	Email	Mobile Number
1	Arun Kumar	arun@example.com	9876543210
2	Priya Sharma	priya@example.com	9123456780

Category

Category ID	Category Name
1	Electronics
2	Books

Product

Product ID	Product Name	Price	Stock	Category ID	Supplier ID
101	Wireless Mouse	599.00	150	1	1
102	Python Programming Book	450.00	80	2	2

Order

Order ID	Customer ID	Order Date	Total Amount	Order Status
5001	1	2026-08-10	1648.00	Delivered

Order Details

Order Detail ID	Order ID	Product ID	Quantity	Unit Price
1	5001	101	2	599.00
2	5001	102	1	450.00

Order 5001 contains two products, so its total is $(2 \times 599) + (1 \times 450) = 1198.00 + 450.00 = 1648.00$. This example shows how one order links to many order detail rows, and how Order Details connects Order and Product.

8. Assumptions Made

1. Every entity uses a system-generated numeric ID as its primary key.
2. A product belongs to exactly one category and one supplier.
3. An order contains at least one product, and a product appears at most once per order (quantity captures multiples).
4. Each order has one payment and one shipment in the basic model.
5. Unit Price is copied from Product at the time of ordering so that future price changes do not affect past orders.
6. A customer may review a product only once.

9. Conclusion

The study identified nine entities: Customer, Category, Supplier, Product, Order, Order Details, Payment, Shipment and Review. For each, the attributes, data types, primary key, candidate keys and foreign keys were defined. Eleven relationships were recorded, including one-to-one (Order-Payment, Order-Shipment), one-to-many (for example Customer-Order, Category-Product) and two many-to-many relationships (Order-Product and Customer-Product) that are resolved through the associative entities Order Details and Review.